

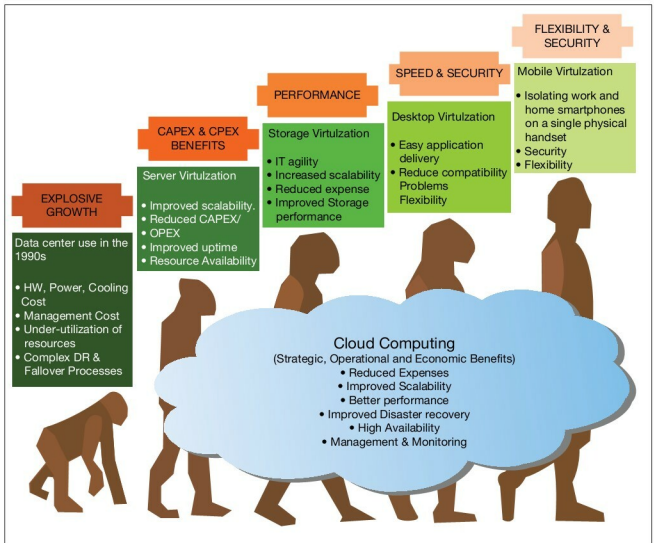


Figure 4: Virtualisation and its benefits

IT challenges that led to virtualisation

- Agile business requirements
- Increased Total Cost of Ownership (TCO)
- Server sprawl
- Low server utilisation
- Complex server-storage migration
- Inefficient server deployment
- The complexity of ensuring high-availability
- The complexity of disaster-recovery
- The need for green IT
- Automating processes
- Policy-driven management

VMware used x86 techniques to address many of these difficulties and convert x86 techniques into a general

purpose, shared hardware infrastructure that offers full isolation, flexibility and a choice of OSs for application environments.

After 2000, there has been a lot of success in the area of virtualisation. In the mid-2000s, both Intel and AMD included hardware assistance to their processor chips making virtualisation software easier, and later hardware changes offered considerable speed improvements.

Subsequently, software vendors have developed virtualisation solutions, and organisations have implemented virtualisation to solve business needs.

The benefits of virtualisation

There has been a natural development towards organisations wanting their own servers, data centres, networks and desktop environments and being able to manage them. The preferred end-state would be an